



BEng, BSc, MEng and MMath Degrees 2023–24

Open Assessment

Department Computer Science

Module DATA: Introduction to Data Science

Title DATA: Introduction to Data Science

Issued Thursday, 07 12 2023

Submission due 12 noon, Thursday, 18 01 2024

Feedback and Marks due Thursday, 22 02 2024

Word Limit All tasks specify strict limits on the length of your answers; text exceeding the limit will not be marked

Instructions:

All students should submit their answers through the electronic submission system: the Department's Teaching Portal by 12 noon, Thursday, 18 01 2024. An assessment submitted after this deadline will be marked initially as if it had been handed in on time, but the Board of Examiners will normally apply a lateness penalty.

Your attention is drawn to the section about Academic Misconduct in your Departmental Handbook.

Any queries you may have on this assessment should be posted on the Discussion Board on the Virtual Learning Environment (VLE) page for the module DATA: Introduction to Data Science in the appropriate discussion area. **No questions will be answered after Thursday, 21 12 2023.**

Note on Academic Integrity

We are treating this online examination as a time-limited open assessment, and you are therefore permitted to refer to written and online materials to aid you in your answers. However, you must ensure that the work you submit is entirely your own, and for the whole time the assessment is live you must not:

- communicate with other students on the topic of this assessment.
- seek advice or contribution from any other third party, including proofreaders, friends, or family members.

We expect, and trust, that all our students will seek to maintain the integrity of the assessment, and of their award, through ensuring that these instructions are strictly followed. Where evidence of academic misconduct is evident this will be addressed in line with the Academic Misconduct Policy and if proven be penalised in line with the appropriate penalty table. Given the nature of these assessments, any collusion identified will normally be treated as cheating/breach of assessment regulations and penalised using the appropriate penalty table (see AM3.3. of the Guide to Assessment).

Background

Business Domain

In the bustling UK financial landscape, a well-established bank provides multiple financial services dedicated to individual customers. These services encompass a comprehensive suite of offerings, including but not limited to managing various types of accounts, facilitating the provision of loans, issuing credit cards tailored to different customer needs, and overseeing a wide range of financial transactions.

Acknowledging the importance of data in driving effective, accurate, fair and transparent decision-making, the CEO of the bank has established the Data Intelligence team, designating you as its leader.

Aim: The Data Intelligence team is expected to implement a data science project and present its main findings to the key stakeholders of the bank. These findings will serve as a cornerstone for improving customer service and inform data-driven decision-making within the organisation, for instance, to perform customer risk assessment and establish a customer retention scheme to reward high-value and loyal customers.

Data Overview

The bank maintains records of its clients, including an identifier, birth number (a number based on the date of birth, assigned by the bank), the residential city of the client, and a list of demographic statistics of the city. The demographics include an identifier of the UK city (a1), city name (a2), region of the city (a3), no. of inhabitants (a4), no. of municipalities with inhabitants <500 (a5), no. of municipalities with inhabitants 500-1999 (a6), no. of municipalities with inhabitants 2000-10000 (a7), no. of municipalities with inhabitants >10000 (a8), no. of cities in the region (a9), ratio of urban inhabitants (a10), average salary (a11), unemployment rate 1995 (a12, a measure of the unemployed labour force in 1995), unemployment rate 1996 (a13, a measure of the unemployed labour force in 1996), no. of entrepreneurs per 1000 inhabitants (a14), no. of committed crimes in 1995 (a15), and no. of committed crimes in 1996 (a16).

Each client has one or more accounts, which serve as the primary repository for their financial activities. The stored details include an account identifier, opted frequency of account statements, and account creation date. These accounts facilitate transactions, standing payment orders, loans, and other financial operations. An account is owned by a single individual as a registered owner (called owner), but may also have another individual as a disponent owner¹ (called disponent). The disposition details are stored with a disposition identifier that links a client (via client identifier) with an account (account identifier) and the type of disposition (whether owner or disponent).

¹ A disponent owner of an account refers to an individual who has the authority to manage or control the funds within an account (i.e., conduct financial transactions, make withdrawals, and manage the assets), even if they are not the primary or sole owner of the account.

A client can perform various types of transactions using their account. Each transaction captures essential details, providing a comprehensive overview of the financial activities within the bank. The recorded details include a transaction identifier that enables tracking and management of individual transactions, an identifier of the account performing the transaction, date of occurrence, type of transaction, nature of financial operation carried out, transaction amount, remaining balance, detailed characterisation of the transaction, bank of the partner involved in the transaction, and account of the partner involved in the transaction. The transaction details form a crucial component of the bank's database, providing critical insights into the financial health and operational dynamics of the institution.

A client can also initiate a permanent standing order using their account. The bank keeps a record of all the initiated orders. Each record holds details related to the processing of payment orders. These include an order identifier enabling their systematic tracking, the identifier of the account that initiated the respective order, the recipient bank involved in the payment order, the recipient account to which the payment is to be made, the amount of the order, and the payment type characterising the nature of the payment.

Furthermore, the bank offers loans to eligible clients based on predetermined criteria. The bank maintains a record of all approved loans, including a loan identifier, an identifier of the account associated with the loan, approval date, amount approved, duration, monthly payments, and loan status.

The bank also issues credit cards to clients, enabling them to make purchases and access credit within predefined limits. The credit card usage is closely linked to the clients' financial behaviour and creditworthiness. In this regard, the bank stores a credit card identifier, a disposition identifier to which the card is issued, its type (gold, classic, or junior), and its issue date.

Tasks

You are required to design and build solutions based on the background scenario discussed (pages 3-4) and the data provided (page 6). There are five specific tasks: Task 1 – Task 5. In particular,

- Task 1: conduct domain analysis;
- Task 2: design the database for the provided scenario and data;
- Task 3: design a research plan to analyse the provided data based on the business domain and needs;
- Task 4: execute the plan and extract actionable knowledge;
- Task 5: conclude the data science project.

A detailed description of each task is provided below.

Your findings and code should be presented as a Jupyter notebook and include the tasks described below. The dataset (*BankRecords.csv*) and a template of the Jupyter notebook with the five tasks are available on VLE (*VLE → DATA → Assessment → Assessment 2023-24 resources*). Further formatting details and word limit per task are given below.

Task 1: Domain Analysis (5 Marks)

Given the business domain and the data overview presented above, provide a brief description of

- the business problem and its significance to the relevant sector;
- the link between the business problem and the field of data science;
- the main areas of investigation;
- potential ideas and solutions.

Word Limit: Your answer to this question must not exceed **300 words**.

Task 2: Database Design (25 Marks)

Having understood the business domain, present a conceptual design in the form of an entity-relationship (ER) model that would be helpful in creating a database for the bank.

The bank data currently exists in the form of a csv file called *BankRecords.csv*, provided on VLE (path given in page 5). This file has all the existing records. The table available in the csv file is unnormalised. The information about its different columns is given in Tables 1 and 2.

Following the standard principles of database normalisation, normalise the given table (*BankRecords.csv*) to a database schema that has minimum redundancies. Then, using the designed schema, create an SQLite database.

Your answer should include the SQL statements needed to accomplish this step. Your submission should also include the created SQLite database file.

Your answer should clearly cover the following:

- Any assumptions you are making about the given scenario;
- The designated keys, existing relationships, and identified functional dependencies;
- The steps followed and justifications for the decisions made.

World Limit: Your answer to this question must not exceed **500 words**. This limit applies only to the explanations. There is no limit on any associated code/SQL statements or figures.

Task 3: Research Design and Implementation (25 Marks)

Using the database designed in Task 2, design and implement **five** potential modelling solutions to achieve the aim of the Data Intelligence team. You need to provide clear justifications about the techniques selected in the context of the 'problem in hand'. Your design must consist of a combination of inferential statistics, supervised learning algorithms, and unsupervised learning algorithms, and include **at least one** of those techniques. Finally, your modelling solutions should be of sufficient complexity, combining information from multiple tables from the database built in Task 2, as appropriate. Your answer should clearly show the queries made to the database. If amendments are made to the database, the commands should be clearly included in your answer.

Your answer should clearly cover the following:

- Any assumptions you are making about the given scenario;
- Any data processing and data integrity steps you would undertake to make the data fit for purpose;
- Which technique(s) you would apply for each solution and why;
- An evaluation of the techniques applied in terms of the accuracy of their results (or any other suitable evaluation measure);
- Algorithmic parameters should be adequately stated and discussed;
- A discussion of ethical considerations arising from the solutions selected.

World Limit: Your answer to this question must not exceed **500 words**. This limit applies only to the explanations. There is no limit on any associated code or figures.

Task 4: Experimental Results and Analysis (25 Marks)

Given the **five** modelling solutions implemented above, analyse, discuss and present your findings to the key stakeholders of the bank.

Your answer should clearly cover the following:

- Present your findings in a clear and concise manner;
- Discuss your results in the context of the selected solution;
- Discuss how these results can help the bank in performing customer risk assessment and establishing customer retention strategies;
- Present the limitations (if any) of your solutions in a clear and concise manner.

World Limit: Your answer to this question must not exceed **500 words**. This limit applies only to the explanations. There is no limit on any associated code or figures.

Task 5: Conclusion (10 Marks)

Given the insights derived from Tasks 1–4, provide a conclusion that clearly covers the following:

- A summary of the main points;
- A discussion of the significance of your results;
- Any recommendation(s) resulting from your analysis;
- Any overall ethical considerations arising from the data analysis of this business domain.

World Limit: Your answer to this question must not exceed **300 words**.

Overall Academic Quality (10 Marks)

10 marks are allocated for the clarity and cohesiveness of your answers (both text and code) across all tasks with appropriate, relevant and effective analysis and presentation of the results.

Deliverables

You should submit the following to the submission point on the teaching portal:

1. the SQLite database produced in Task 2;
2. the completed Jupyter notebook (both .ipynb and HTML files) that also includes the SQL statements (Task 2), the research design and its implementation (Task 3), and the analysis and presentation of your results (Task 4);
3. any figures or diagrams that are included in your answers in the Jupyter notebook.

Table 1: Description of columns in BankRecords

Attribute name	Description	Remark
account_id	identifier of the account	
frequency	frequency of statement issuance	<i>Monthly, Weekly, Per transaction</i>
creation_date	date of creating of the account	in the form YYMMDD
loan_id	loan identifier	
loan_date	date when the loan was granted	in the form YYMMDD
loan_amount	amount of money	
loan_duration	duration of the loan	
loan_payments	monthly payments	
loan_status	status of paying off the loan	'A' (contract finished, no problems), 'B' (contract finished, loan not paid), 'C' (running contract, OK so far), 'D' (running contract, client in debt)
order_id	permanent order identifier	
bank_to	bank of the recipient	each bank has unique two-letter code
account_to	account of the recipient	
order_amount	debited amount	
payment_type	characterization of the payment	<i>Insurance payment, Household, Leasing, Loan payment</i>
trans_id	transaction identifier	
trans_date	date of transaction	in the form YYMMDD
trans_type	+/- transaction	<i>Credit, Withdrawal, Withdrawal in cash</i>
operation	mode of transaction	<i>Credit card withdrawal, Credit in cash, Collection from another bank, Withdrawal in cash, Remittance to another bank</i>
trans_amount	amount of money	
balance	balance after transaction	
trans_detail	detailed characterisation of the transaction	<i>Insurance payment, Payment for statement, Interest credited, Sanction interest if negative balance, Household, Old-age pension, Loan payment</i>
partner_bank	bank of the partner	each bank has unique two-letter code
partner_account	account of the partner	
birth_number	identification of client	the number is in the form YYMMDD for men, and in the form YYMM+50DD for women, where YYMMDD is the date of birth
disp_id	disposition identifier	
client_id	client identifier	
disp_type	type of disposition (owner/user)	only owner can issue permanent orders and ask for a loan
card_id	card identifier	
card_type	type of card	<i>gold, classic, junior</i>
card_issued	issue date	in the form YYMMDD

Table 2: Description of columns in BankRecords

Note: The attribute values a1-a16 are hypothetical

Attribute name	Description	Remark
a1	city code	
a2	city name	
a3	region	
a4	no. of inhabitants	
a5	no. of municipalities with inhabitants < 500	
a6	no. of municipalities with inhabitants 500-2000	
a7	no. of municipalities with inhabitants 2000-10000	
a8	no. of municipalities with inhabitants >10000	
a9	no. of areas	
a10	ratio of urban inhabitants	
a11	average salary	
a12	unemployment rate in 1995	
a13	unemployment rate in 1996	
a14	no. of entrepreneurs per 1000 inhabitants	
a15	no. of committed crimes in 1995	
a16	no. of committed crimes in 1996	

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